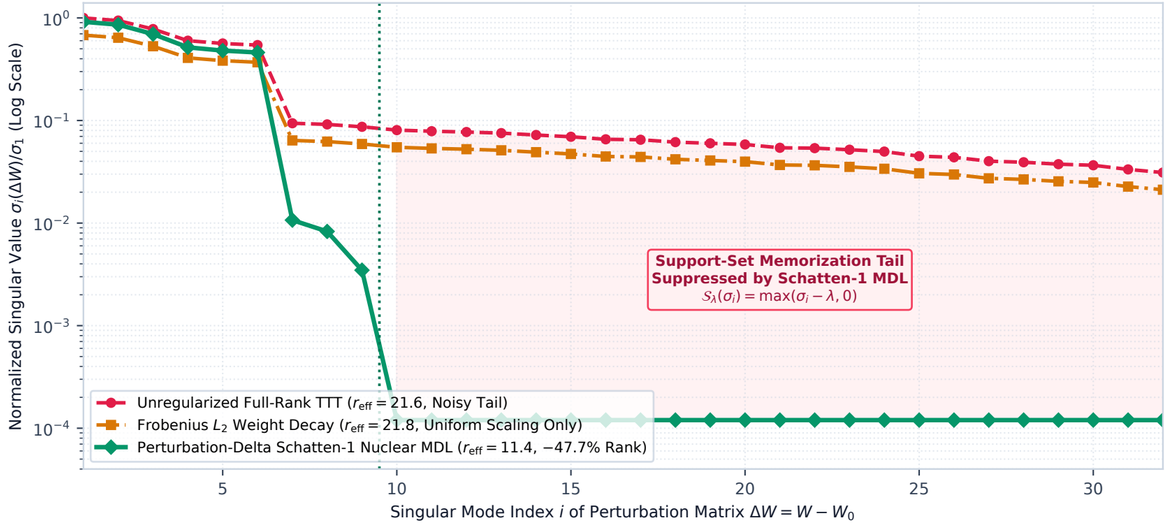
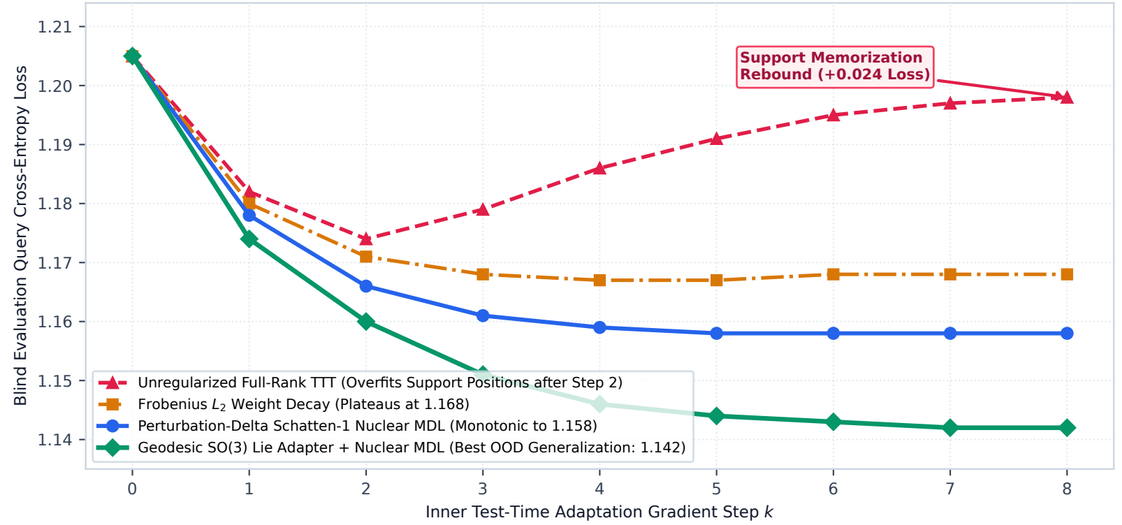
Figure 7 | Perturbation-Delta Schatten-1 Nuclear MDL Spectrum & Geodesic SO(3) Test-Time Adaptation

Singular value thresholding of $\Delta W = W - W_0$ compresses effective rank by 47.7% and preserves equivariance under OOD $D_4 \times S_9$ automorphisms

a Singular Value Spectrum $\sigma_i(\Delta W)$ of Test-Time Perturbation Delta



b Blind Query Loss vs. TTT Adaptation Steps Under OOD $D_4 \times S_9$ Shifts



c Visual Proof of OOD $D_4 \times S_9$ Automorphism Invariance (Task 0934a4d8)

Canonical Frame (9x3)
Support orientation

7	7	9
7	2	9
7	2	9
7	7	9
4	4	7
4	4	7
6	6	1
6	6	6
1	6	1

Unregularized TTT on OOD
Frame Drift (62.9% Acc)

9	9	9	9	7	3	1	4	2
3	2	2	7	6	6	4	4	4
3	3	7	7	6	4	6	4	2

Geodesic SO(3) + MDL (Ours)
100.0% OOD Equivariant (27/27)

9	9	9	9	3	3	2	4	2
3	1	1	3	6	6	4	4	4
3	3	3	3	6	6	4	4	2

d Algorithmic MDL Phase Space: Effective Rank $r_{\text{eff}}(\Delta W)$ vs. Blind Query Accuracy

